Konn3ct Edge Blueprint

Volume 7

Quality Assurance, Testing and Release Governance

Full Draft with Diagram Placeholders

Prepared for internal review, sprint governance and production release discipline

Strategic Theme

No sprint should move to human QA or production release until the independent AI QA engine has tested it against approved SLA rules and produced release-readiness evidence.

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7.1 Purpose of Volume 7

Volume 7 defines the quality assurance, testing and release governance system required to protect Konn3ct before every production release. It converts product ambition, technical architecture and commercial expectations into a disciplined quality operating model.

Konn3ct is not only a meeting tool. It is a platform for live meetings, webinars, paid events, recordings, paid replays, AI summaries, mobile access, customer support, Odoo/CRM integration, payments and future international growth. Each sprint must therefore be validated against real business risk, not only against basic functional checks.

Quality purpose: Konn3ct should release based on evidence, SLA compliance, AI-led automated testing, human validation and formal governance approval.

- Define how QA protects customer trust, revenue and product reputation.
- Create an independent AI QA engine that tests every sprint before human QA.
- Establish intelligent and self-learning SLA management.
- Make OS, device, browser and version coverage a formal certification requirement.
- Ensure failed components produce actionable recommendations for developers and developer engines.
- Support flexible test scenarios for meetings, webinars and all associated subsystems.

7.2 Quality Assurance Vision for Konn3ct

The QA vision is to make Konn3ct a platform that customers can trust for serious meetings, paid events, institutional sessions, online training, faith-based gatherings, enterprise collaboration and large community events.

QA must be treated as a business trust function. A failed meeting join can reduce adoption. A broken paid access rule can damage revenue. A public recording link can destroy trust. A mobile crash can block user growth. A poor webinar experience can damage the brand.

- QA protects Free Users by ensuring first-time users can register, join and understand the platform without friction.
- QA protects Paid Users by ensuring payments, access, replays, support and reliability work as promised.
- QA protects revenue by validating paid event, paid replay, subscription and renewal workflows.
- QA protects reputation by preventing obvious defects from reaching customers.
- QA protects development speed by catching defects early and giving clear recommendations.

7.3 QA Governance Principles

Konn3ct QA should be governed by principles that apply to every sprint and every release.

Principle	Meaning	Expected Practice
Test before confidence	No team should assume a build is ready because it looks ready.	Every release must produce test evidence.
AI first, human accountable	AI testing should run before human QA, but humans remain responsible for final judgment.	AI QA -> human QA -> product sign-off -> technical approval.
SLA-driven quality	Quality must be measured against approved thresholds.	SLA library controls pass, hold and fail decisions.
Compatibility by evidence	Browsers, devices and OS versions must be certified, not guessed.	Maintain living compatibility matrix.
Customer feedback learns	Support issues must become QA learning signals.	Repeated defects create regression tests.
Failed tests advise developers	A failed test must recommend what to inspect or fix.	Generate component, root-cause and fix-direction output.
Block poor releases	Critical failures must stop release movement.	Enforce go/no-go criteria.

7.4 End-to-End QA Operating Model

The QA operating model begins before coding starts. Requirements, user stories and acceptance criteria must be reviewed for testability. The AI QA engine should generate test cases and expected outcomes from those requirements. Development then produces a sprint build, which is tested automatically against the approved SLA library before human QA begins.

Human QA then validates user experience, edge cases, real-life workflows, webinar behavior, meeting quality, mobile usability and release evidence. Product owner and technical lead approval are required before production deployment. After release, monitoring and support feedback continue the QA cycle.

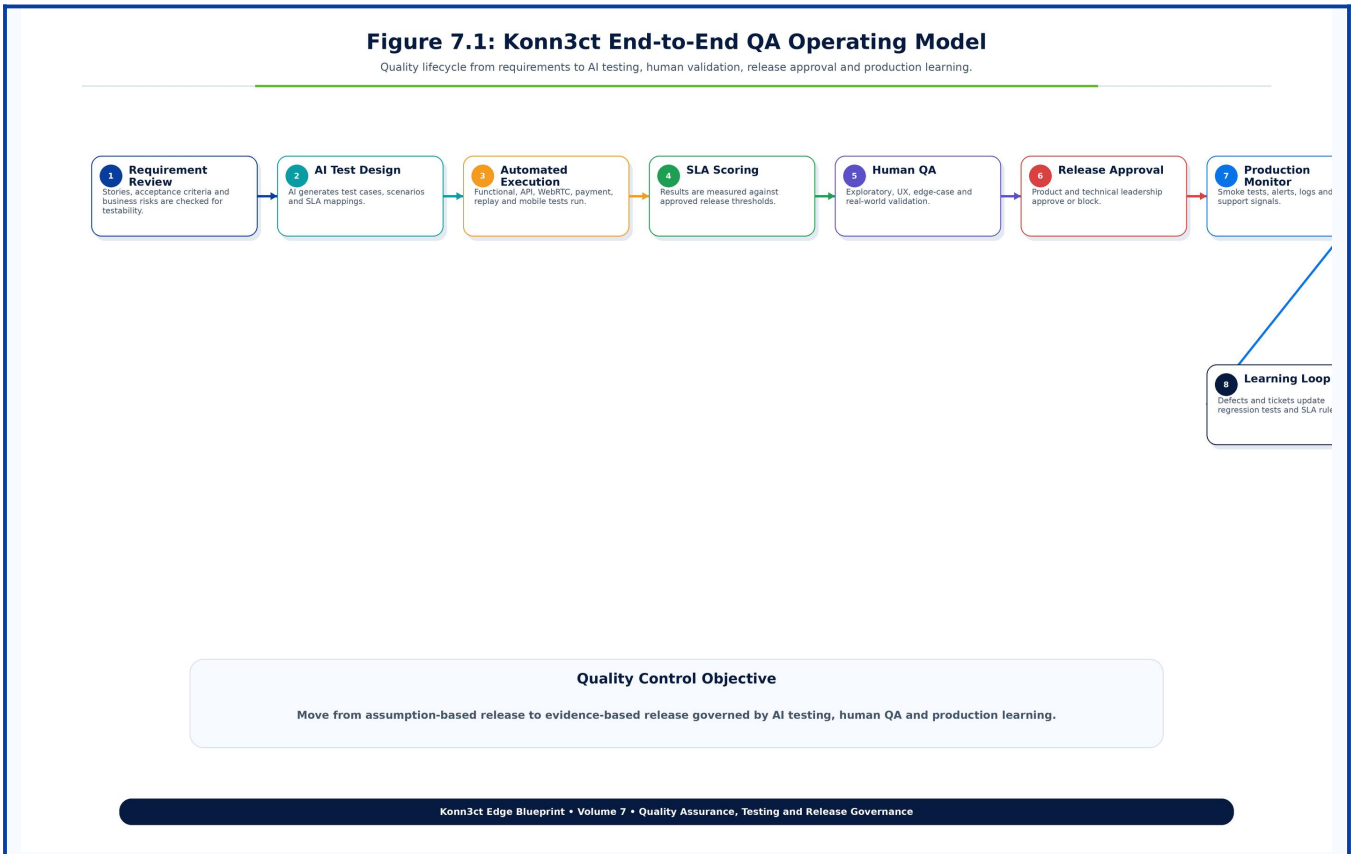


Figure 7.1: Konn3ct End-to-End QA Operating Model

Show requirement review, AI test generation, automated testing, SLA scoring, human QA, release approval, production monitoring and customer feedback learning.

1. Requirement review and testability analysis.
2. AI-generated test plan and SLA mapping.
3. Automated sprint test execution.
4. Release readiness scoring and defect recommendation.
5. Human QA validation and exploratory testing.
6. Product owner and technical lead approval.
7. Production release, smoke test and monitoring.
8. Support feedback and regression learning.

7.5 Independent AI Test and Quality Assurance Engine

The Independent AI Test and Quality Assurance Engine is a dedicated testing layer that evaluates every sprint before it goes to human QA. Its purpose is to test faster, broader and more consistently than manual QA alone.

The engine should understand sprint requirements, user stories, acceptance criteria, code changes, known defects, support complaints, logs, test history and SLA rules. It should generate test cases, run automated tests, classify failures, score release readiness and recommend whether a build should pass, hold or fail.

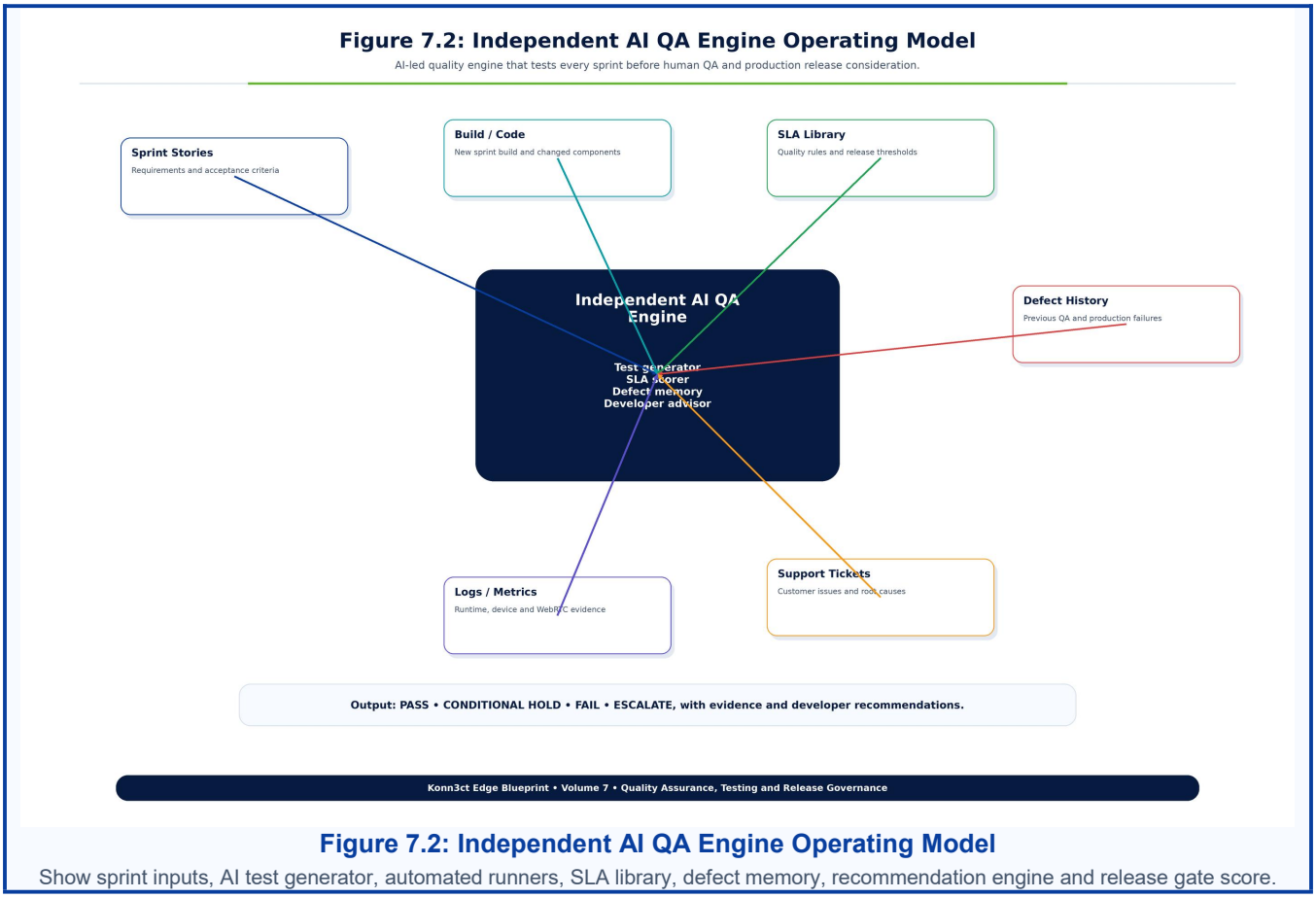


Figure 7.2: Independent AI QA Engine Operating Model

Show sprint inputs, AI test generator, automated runners, SLA library, defect memory, recommendation engine and release gate score.

AI QA Capability	Description	Output
Requirement interpretation	Reads acceptance criteria and expected business outcome.	Testable scenarios and edge cases.
Test generation	Creates functional, API, UI, WebRTC, mobile, payment, replay and AI tests.	Automated and manual test candidates.
Execution orchestration	Runs tests or triggers appropriate automated test suites.	Pass/fail evidence and logs.
Defect reasoning	Connects failure to likely component, root cause and risk.	Developer advisory report.
SLA scoring	Scores build against approved quality thresholds.	Pass, conditional hold or fail.
Learning loop	Learns from approved support tickets, incidents and regression history.	New tests and SLA update suggestions.

7.6 Intelligent SLA Library and Self-Learning Quality Rules

Konn3ct must maintain an Intelligent SLA Library. This library should define what quality means for each major feature, workflow, device environment and release condition. The SLA library must be machine-readable enough for the AI QA engine and understandable enough for QA, product, development and leadership teams.

The SLA library should not be static. It must learn from failed tests, production incidents, customer complaints, support tickets, app crashes, payment errors, replay access issues and repeated defects. However, the system should recommend SLA updates; it should not silently change critical release thresholds without human approval.

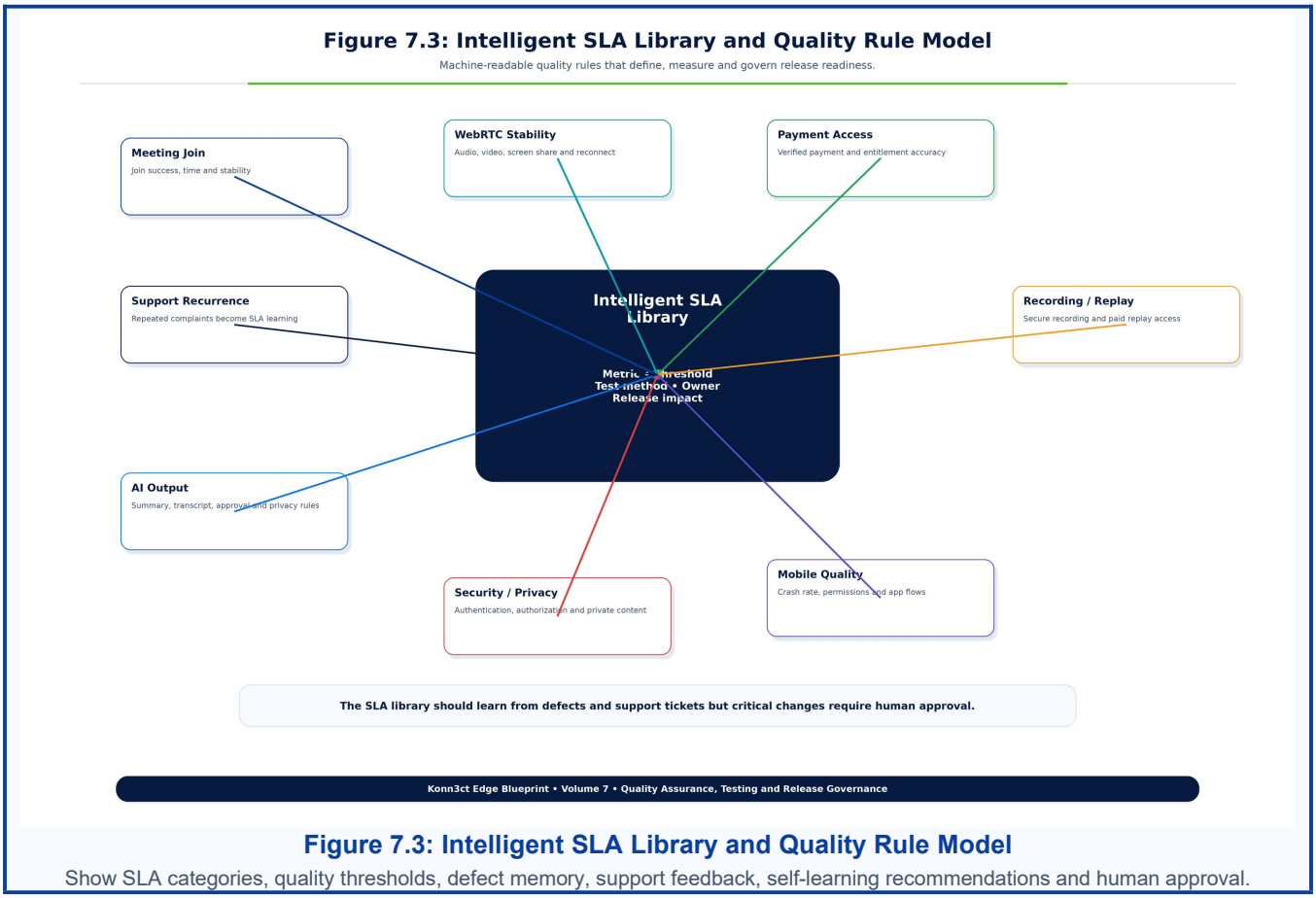


Figure 7.3: Intelligent SLA Library and Quality Rule Model

Show SLA categories, quality thresholds, defect memory, support feedback, self-learning recommendations and human approval.

SLA Domain	Example Quality Rule	Learning Inputs
Meeting join	Supported users must be able to join within defined timing and reliability thresholds.	Join failures, browser issues, support tickets.
WebRTC quality	Audio/video, screen share and reconnection must pass defined test thresholds.	WebRTC stats, diagnostics overlay, user complaints.
Payment and access	Paid access must activate only after verified payment confirmation.	Payment tickets, webhook failures, access complaints.
Recording and replay	Recordings must be generated, secured, published and accessed according to policy.	Replay issues, storage errors, privacy findings.
Mobile	Supported devices and OS versions must pass critical mobile journeys.	Crash reports, app feedback, device test results.
AI features	AI outputs must be generated securely, within expected time, with human approval where required.	AI job failures, user feedback, support tickets.
Security	Authentication, authorization and private content access must pass release-blocking checks.	Security testing, audit findings, incidents.

7.7 SLA Management, Ownership and Approval Governance

SLA management should be a governed process. SLAs should be owned by QA leadership with input from product, technical leads, support, DevOps and business leadership. Every major SLA must have an owner, a measurement method, a testing method and a release impact.

The self-learning SLA system may recommend new thresholds, new regression tests or new risk ratings. Those recommendations should be reviewed and approved before becoming active release-blocking rules.



Figure 7.4: SLA Management and Self-Learning Approval Workflow

Show SLA proposal, evidence source, AI recommendation, QA review, product/technical approval, SLA activation and audit log.

SLA Governance Item	Required Decision	Owner
New SLA rule	Should the rule be added to release gate?	QA Lead + Product Owner
Critical threshold change	Will this block production releases?	QA Lead + Technical Lead
Support-driven rule	Is the customer issue recurring enough to become a regression rule?	Support Lead + QA Lead
Device/browser certification rule	Is the environment supported or high-risk?	QA Lead + Engineering Lead
AI-generated recommendation	Is the suggested rule technically valid and fair?	QA Lead + Technical Lead

7.8 Sprint Quality Gate Model

Every sprint must pass through a quality gate before human QA and before production release consideration. The sprint quality gate is the formal checkpoint where the AI QA engine evaluates build readiness against the SLA library.

The quality gate should produce a score, failed component list, recommendation report and release decision. The decision should be Pass, Conditional Hold, Fail or Escalate.

DIAGRAM PLACEHOLDER

Figure 7.5: Sprint Quality Gate and Release Readiness Flow

Show sprint build, automated tests, SLA checks, failed component advisory, quality score, pass/hold/fail decision and human QA handoff.

Gate Decision	Meaning	Required Action
Pass	Critical tests passed and evidence is complete.	Proceed to human QA.
Conditional Hold	Non-critical defects or incomplete evidence exist.	Fix or formally accept risk before release approval.
Fail	Critical SLA or core user journey failed.	Block release and return to development.

Gate Decision	Meaning	Required Action
Escalate	Severe security, payment, recording, privacy or meeting failure found.	Immediate product and technical review.

7.9 Human QA Review and Validation Model

Human QA remains essential even with AI testing. Human testers validate user experience, visual quality, edge cases, emotional trust, customer confusion, language clarity, host confidence, webinar control usability and unusual real-world behavior.

Human QA should focus on areas where judgment matters: meeting flow, host confidence, participant clarity, mobile usability, dashboard usefulness, support recovery, campaign-triggered flows, paid event clarity and release readiness documentation.

- Confirm that automated test results match observable product behavior.
- Perform exploratory testing around high-risk workflows.
- Validate user experience across web, mobile and desktop layouts.
- Test real-world webinar and meeting behaviors that may not be captured by static automation.
- Confirm that critical defects have clear evidence, ownership and retest results.
- Prepare release readiness summary for product and technical approval.

7.10 Release Governance and Approval Workflow

Production release should require clear governance. No individual developer, product owner or QA tester should unilaterally release a build that has unresolved critical risks. Release should follow a structured approval chain.

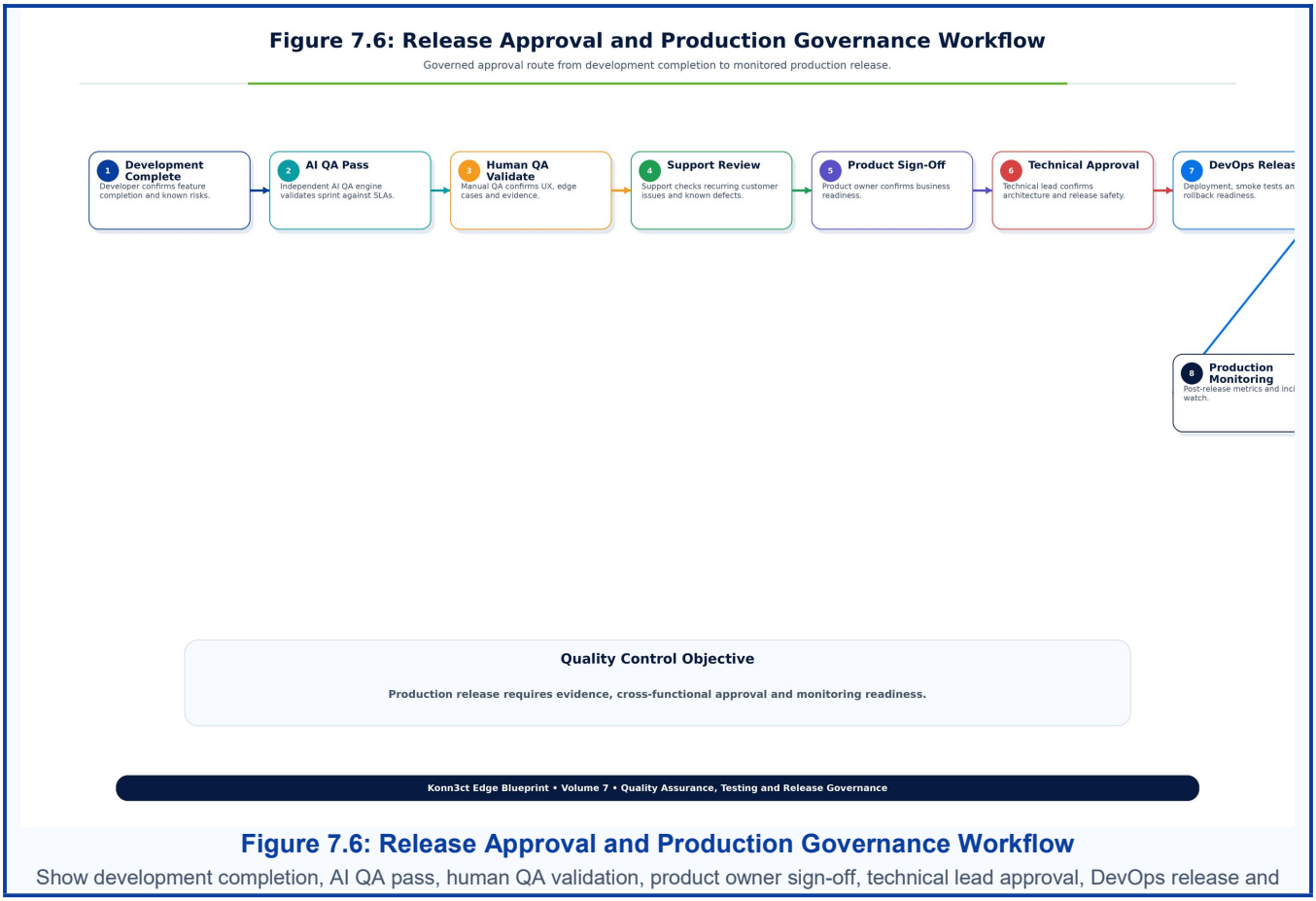


Figure 7.6: Release Approval and Production Governance Workflow

Show development completion, AI QA pass, human QA validation, product owner sign-off, technical lead approval, DevOps release and

post-release monitoring.

1. Developer confirms implementation and known risks.
2. AI QA engine runs automated sprint gate.
3. QA validates failed and passed cases.
4. Support checks known recurring customer issues that may be affected.
5. Product owner validates business readiness.
6. Technical lead validates technical readiness.
7. DevOps confirms deployment and rollback readiness.
8. Production release is monitored with post-release smoke checks.

7.11 Functional Testing Framework

Functional testing validates that core Konn3ct workflows perform as expected. Functional testing should cover both happy paths and realistic failure paths.

Functional Area	Test Focus	Expected Outcome
Registration and login	Signup, login, password reset and account status.	Users can access platform securely.
Scheduling	Meeting creation, event creation, reminders and calendar logic.	Hosts can plan sessions without confusion.
Meeting join	Participant and host join flows.	Users enter the right session with correct role.
Host controls	Mute, remove, lock, admit, spotlight, pin, screen share and recording controls.	Hosts remain in control.
Chat and Q&A	Messaging, moderation, webinar questions and visibility.	Communication works without abuse or confusion.
Dashboards	Host, participant, global admin and support dashboards.	Each role sees accurate and useful information.
Notifications	Email, WhatsApp, push and in-app reminders.	Relevant messages reach users at the right time.

7.12 WebRTC and Meeting Quality Testing

WebRTC and meeting quality testing is one of the most critical QA areas for Konn3ct. Users judge the platform largely by whether they can join, hear, see, share, reconnect and complete the meeting without disruption.

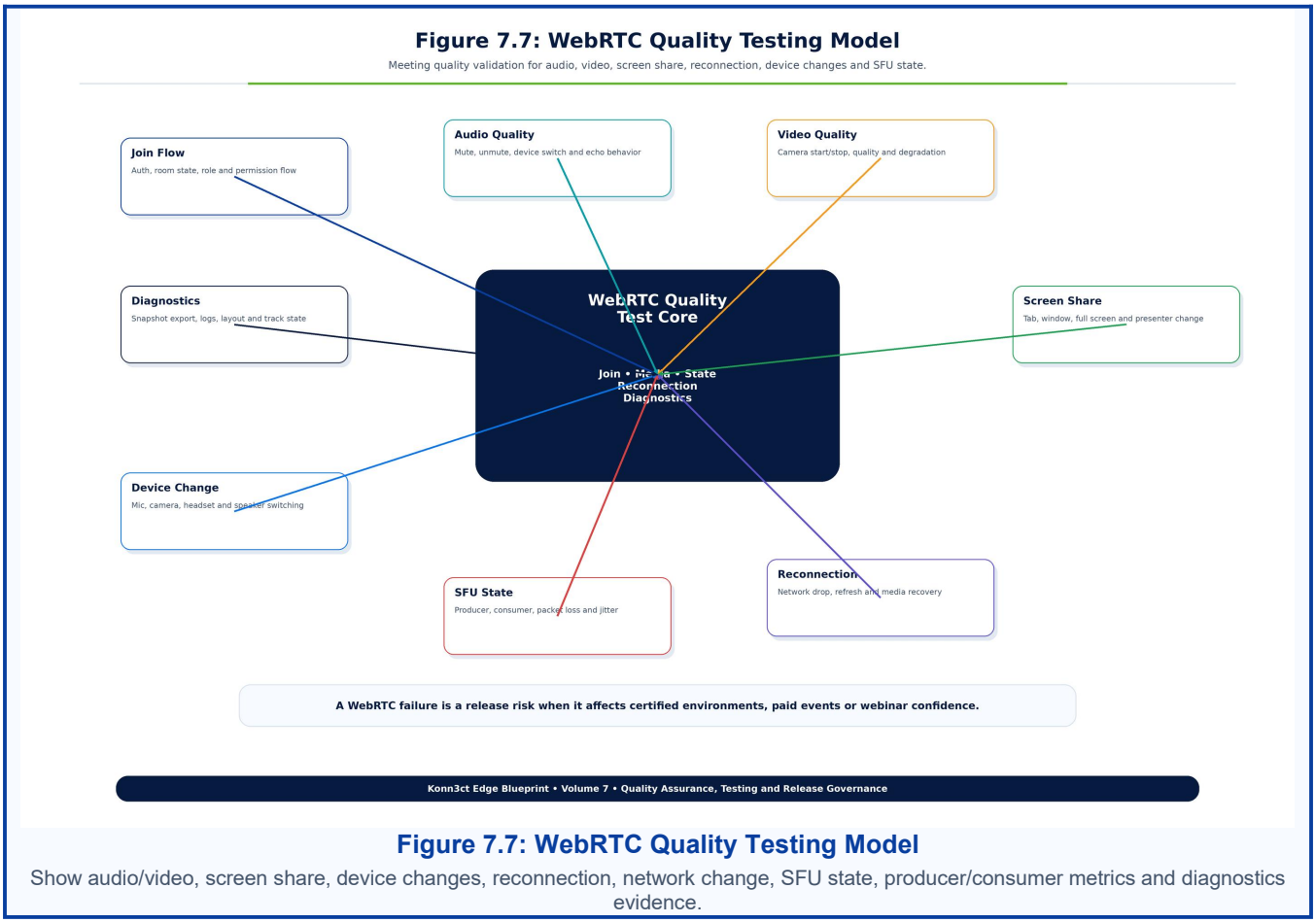


Figure 7.7: WebRTC Quality Testing Model

Show audio/video, screen share, device changes, reconnection, network change, SFU state, producer/consumer metrics and diagnostics evidence.

- Audio join, mute, unmute and device switching.
- Video start, stop, camera change and low-bandwidth behavior.
- Screen share start, stop, presenter change and layout priority.
- Reconnection after network interruption or browser refresh.
- Producer and consumer state validation.
- Room state consistency for host, participant, presenter and moderator.
- WebRTC diagnostics capture for failed sessions.
- SFU load, bandwidth and media quality monitoring under realistic usage.

7.13 Webinar and Large Event Testing

Webinars and large events must be tested separately from normal meetings because the risk profile is different. Webinar users expect controlled roles, reliable presenter experience, audience management, Q&A, chat moderation, attendance tracking, recording and post-event follow-up.

Webinar Area	Test Scenarios	Quality Expectation
Role management	Host, presenter, moderator and attendee changes.	Role changes do not disrupt live event.
Waiting room	Admission, rejection, late join and lock behavior.	Audience flow remains controlled.
Audience controls	Mute audience, disable camera, restrict chat and manage Q&A.	Host retains authority.
Large attendance	Scale tests with different attendee volumes.	Session remains stable under target capacity.
Recording and replay	Record live webinar and publish replay.	Post-event content is available and secure.
Follow-up	Attendance sync, lead capture and campaign triggers.	Business workflow continues after event.

7.14 Flexible Meeting, Webinar and Subsystem Test Scenario Engine

Konn3ct QA must support flexible scenario-based testing. Real users do not all behave the same way. A webinar may include late joiners, failed payments, mobile participants, poor networks, presenter changes, Q&A surges, screen share issues, recording problems and post-event replay purchases.

The flexible scenario engine should allow QA and the AI QA engine to combine variables and generate realistic test cases for meetings, webinars and related subsystems.

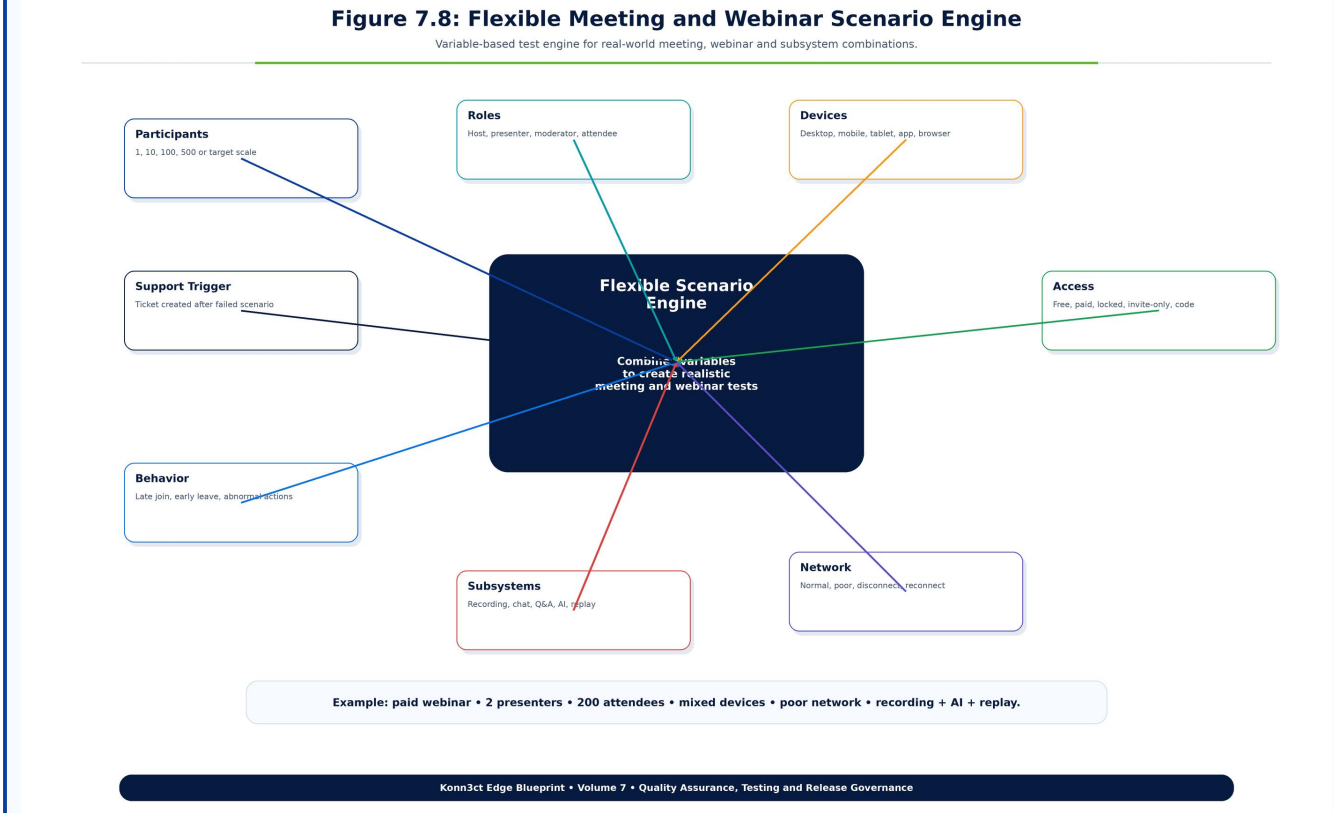


Figure 7.8: Flexible Meeting and Webinar Scenario Engine

Show variable-based test generation across participants, roles, devices, payments, recording, AI, network, chat, Q&A, replay and support tickets.

Scenario Variable	Example Values	Subsystems Affected
Participants	1, 10, 100, 500, target webinar size.	Meeting engine, SFU, dashboards, attendance.
Roles	Host, co-host, presenter, moderator, attendee.	Permissions, layouts, controls, Q&A.
Access type	Free, paid, invite-only, access code, locked meeting.	Payments, entitlements, admission.
Device mix	Desktop, Android, iPhone, tablet, mobile web.	UI, media, performance, crash reporting.
Network condition	Normal, poor, disconnect, reconnect.	WebRTC, reconnection, quality diagnostics.
Subsystems enabled	Recording, replay, AI summary, chat, Q&A, screen share.	Storage, AI jobs, messaging, layout.
Abnormal behavior	Late join, early leave, repeated join, failed payment, spam chat.	Support, audit, abuse prevention, SLA learning.

7.15 Paid Event and Payment Testing

Paid event and payment testing protects revenue and customer trust. Every payment-related path should be tested for success, failure, delay, duplicate confirmation, refund, support override and entitlement accuracy.

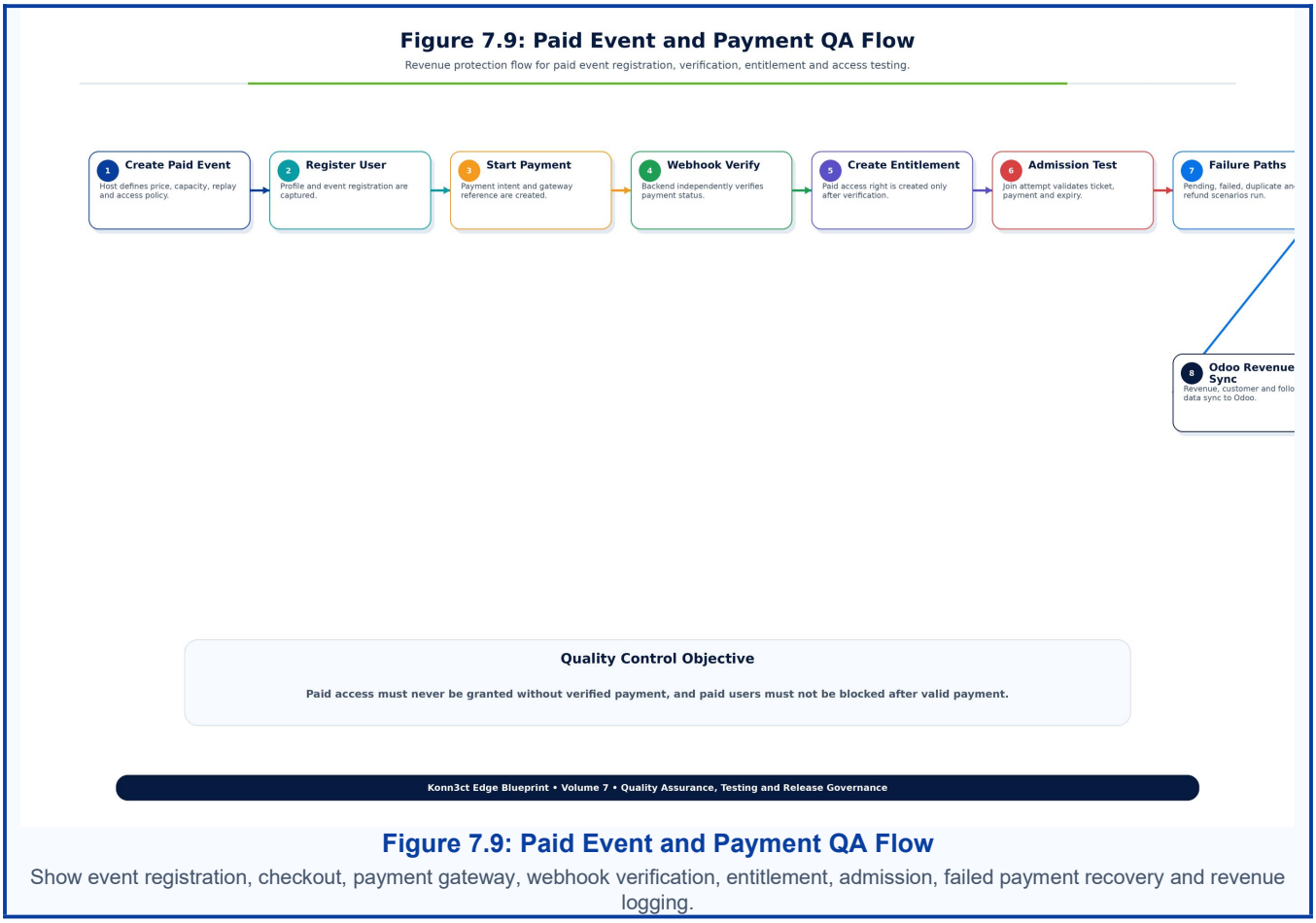


Figure 7.9: Paid Event and Payment QA Flow

Show event registration, checkout, payment gateway, webhook verification, entitlement, admission, failed payment recovery and revenue logging.

- Paid event registration and payment initiation.
- Successful payment webhook and entitlement creation.
- Pending payment and delayed confirmation.
- Failed payment and recovery messaging.
- Duplicate payment detection and support handling.
- Refund and entitlement reversal where policy requires.
- Manual support override with audit log.
- Revenue logging and Odoo synchronization.

7.16 Recording, Replay and Paid Replay Testing

Recordings and replays are strategic assets. QA must ensure that recordings are created reliably, processed correctly, stored securely, published intentionally and accessed according to privacy, subscription and payment rules.

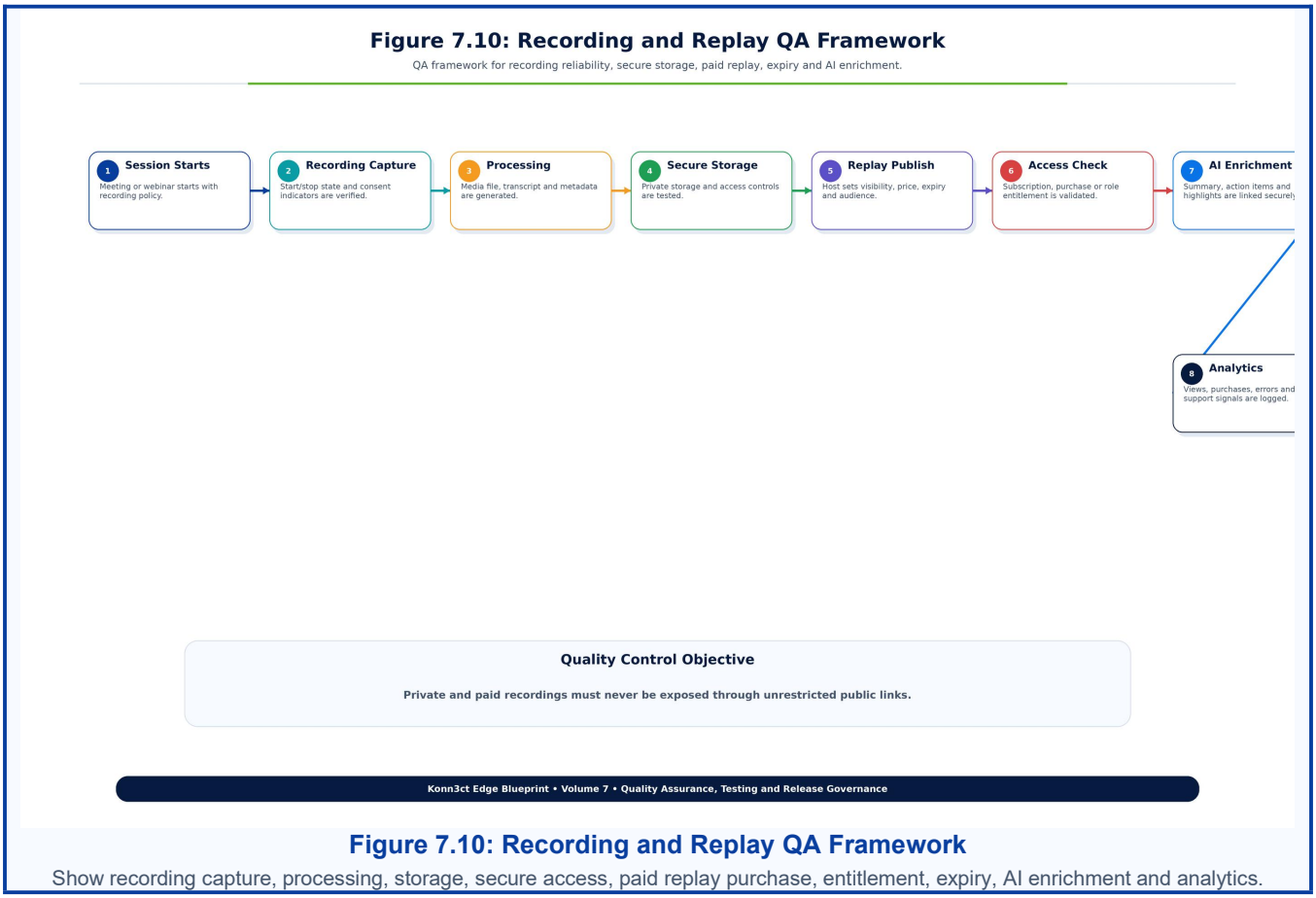


Figure 7.10: Recording and Replay QA Framework

Show recording capture, processing, storage, secure access, paid replay purchase, entitlement, expiry, AI enrichment and analytics.

Replay Test Area	What to Test	Release Risk
Recording creation	Start, stop and recording state across meetings/webinars.	High
Processing	File readiness, failure recovery and metadata.	High
Privacy	Private and paid recordings must not be public unrestricted links.	Critical
Paid replay	Purchase, entitlement, playback and expiry.	High
AI enrichment	Transcript, summary and highlights tied to correct access.	Medium
Storage lifecycle	Retention, expiry, archive and deletion policy.	Medium

7.17 AI Feature Testing

AI features should be tested for functionality, accuracy, privacy, speed, cost and governance. AI outputs should improve customer value without creating confusion, exposing private data or sending unapproved content.

- Meeting summary generation and timing.
- Transcript availability and access control.
- Action item extraction and host approval.
- AI Q&A and answer relevance.
- Replay highlights and content boundaries.
- AI support suggestions and escalation rules.
- AI campaign drafts and human approval workflow.
- AI QA recommendations and developer advisory quality.
- Privacy rules for recordings, transcripts and summaries.
- Cost tracking for repeated or large AI jobs.

7.18 Mobile App QA Framework

Mobile QA must cover Android and iOS as primary channels. The mobile app must support registration, login, meeting join, push notifications, paid access, replay access, AI output access, crash reporting and support escalation.

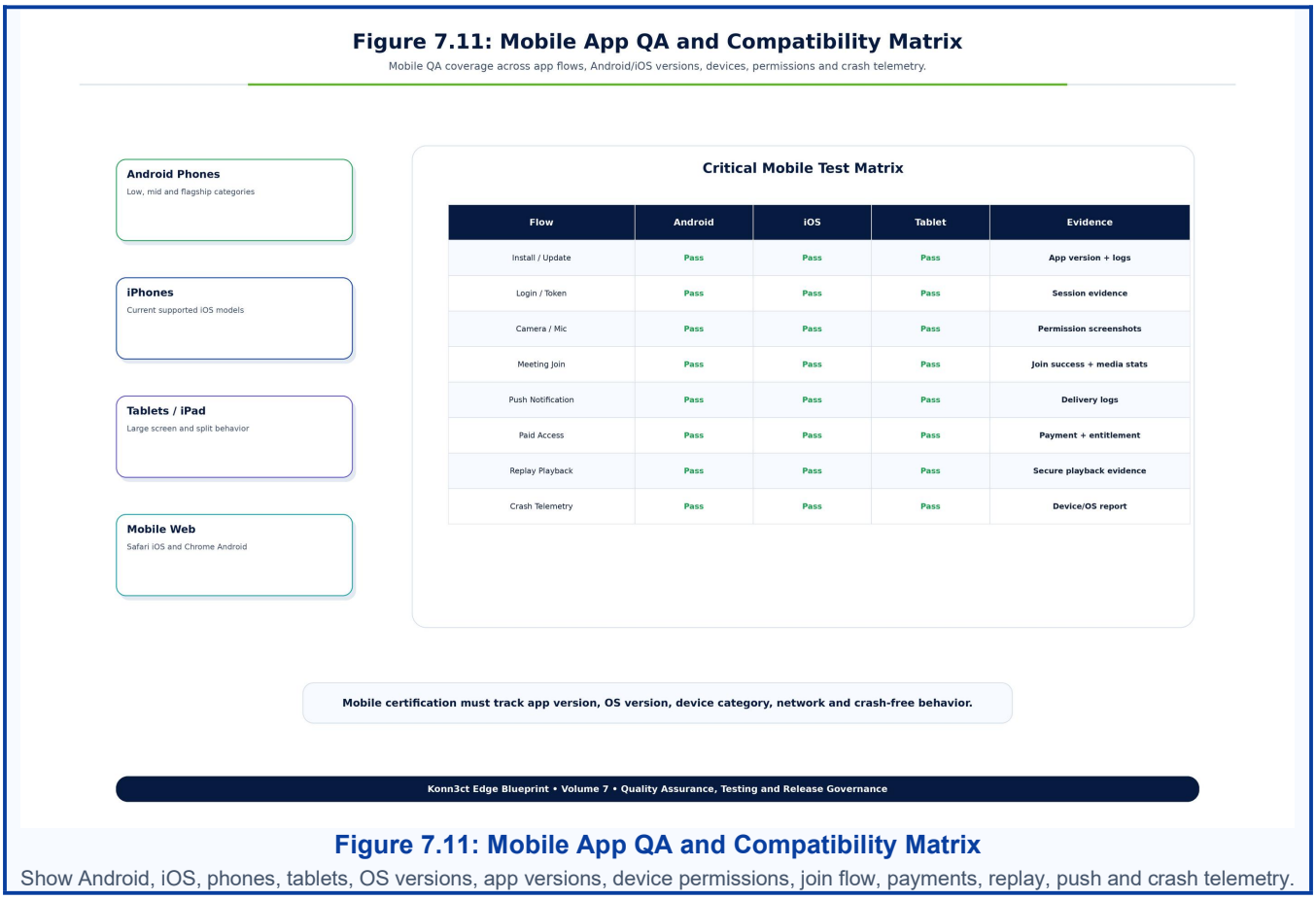


Figure 7.11: Mobile App QA and Compatibility Matrix

Show Android, iOS, phones, tablets, OS versions, app versions, device permissions, join flow, payments, replay, push and crash telemetry.

Mobile Test Area	Required Tests	Evidence
Installation and update	Fresh install, upgrade, app version compatibility.	Install pass/fail and device logs.
Login	Account login, token renewal, logout.	Session and auth test evidence.
Permissions	Camera, microphone, speaker, notification permissions.	Permission behavior screenshots.
Meeting join	Deep link, app join, reconnect, device switch.	Join result and media diagnostics.
Payments	Paid event and replay access.	Payment and entitlement records.
Replay	Replay list, streaming, paid/private access.	Playback result and access rule evidence.
Crash reporting	Crash-free sessions by device and OS.	Crash analytics report.

7.19 Web, Desktop and Browser Compatibility Testing

Browser compatibility testing must verify the Konn3ct experience across supported browsers and versions. The QA process should focus especially on browsers still supported by their vendors and widely used by target customers.

The supported browser matrix should be reviewed regularly because vendor support changes. The QA team should classify browsers as Certified, Supported with Risk, Monitoring Only or Unsupported.

- Chrome and Chromium-based browser behavior.

- Microsoft Edge support and enterprise environment behavior.
- Firefox support for media, permissions and screen sharing.
- Safari support on macOS and iOS where applicable.
- Browser permission prompts for camera, microphone and screen share.
- Layout responsiveness across desktop and laptop screen sizes.
- Dashboard, payment, replay and AI output rendering.
- Browser updates that affect WebRTC, storage, permissions or security.

7.20 OEM-Supported OS, Device and Browser Certification Matrix

Konn3ct QA must test operating systems, device types, browsers and versions that are still supported by original equipment manufacturers or official software vendors. This protects the platform from spending too much effort on obsolete environments while ensuring that current user environments are properly certified.

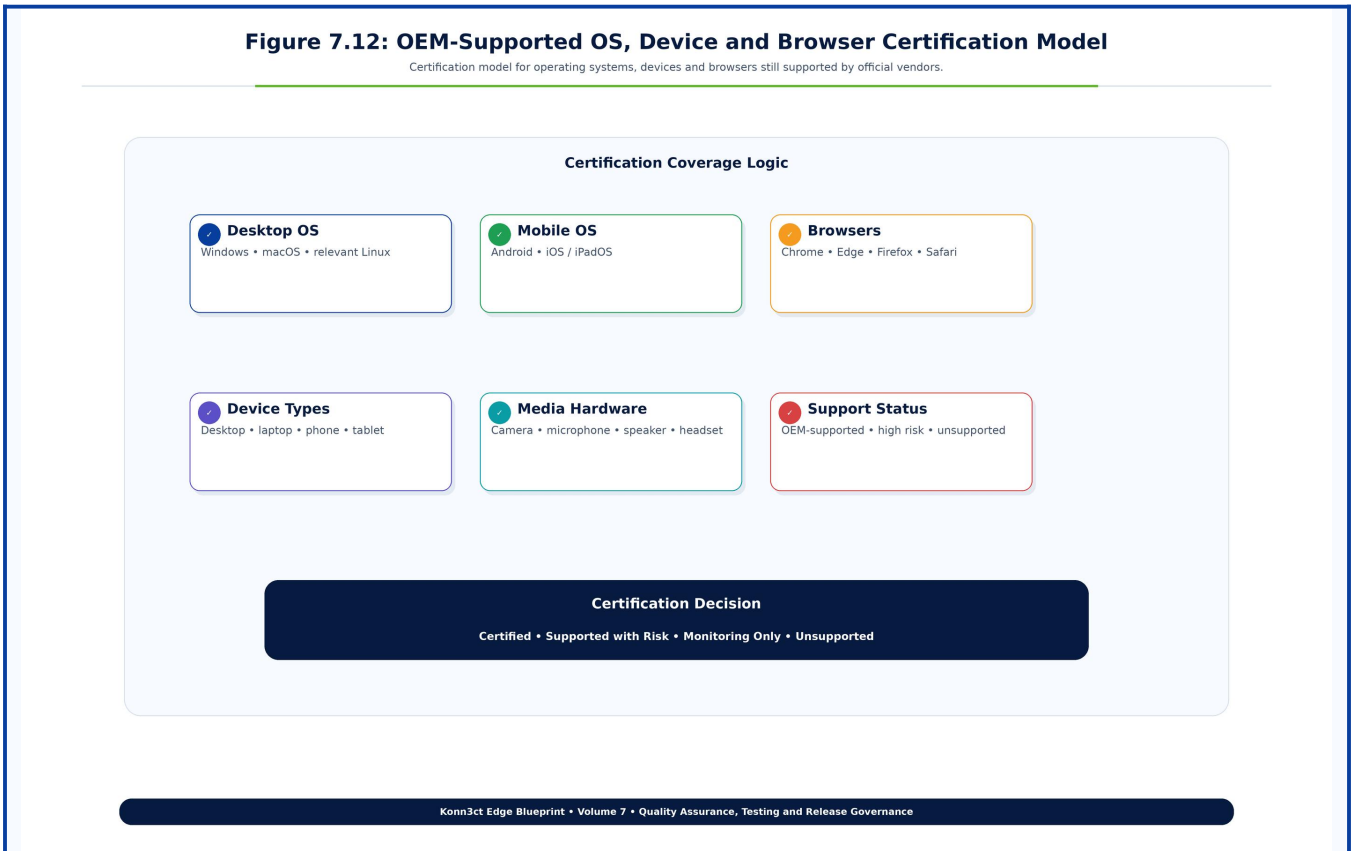


Figure 7.12: OEM-Supported OS, Device and Browser Certification Model

Show operating systems, device types, browsers, versions, OEM support status, certification level, risk score and retest cadence.

Environment Category	Certification Coverage	Certification Rule
Desktop OS	Supported Windows, macOS and relevant Linux distributions.	Test versions still supported by their vendors and relevant to customer base.
Mobile OS	Supported Android and iOS versions.	Prioritize currently supported versions and high-usage device categories.
Desktop browsers	Chrome, Edge, Firefox and Safari versions still supported by vendors.	Certify core meeting, screen share, payment, replay and dashboard flows.
Mobile browsers	Chrome Android, Safari iOS and key mobile browser variants.	Certify login, join, payment and replay flows.
Device types	Desktop, laptop, Android phone, iPhone, tablet and iPad.	Test device categories, not every physical model, unless high-risk.
Peripherals	Webcams, microphones, speakers, headsets and Bluetooth	Test representative hardware and known

Environment Category	Certification Coverage	Certification Rule
	audio.	customer environments.

7.21 Dynamic Compatibility Matrix and Environment Risk Scoring

The compatibility matrix should be a living QA asset. It should track each tested environment, vendor support status, last test date, test result, known issues, recurrence frequency and risk score.

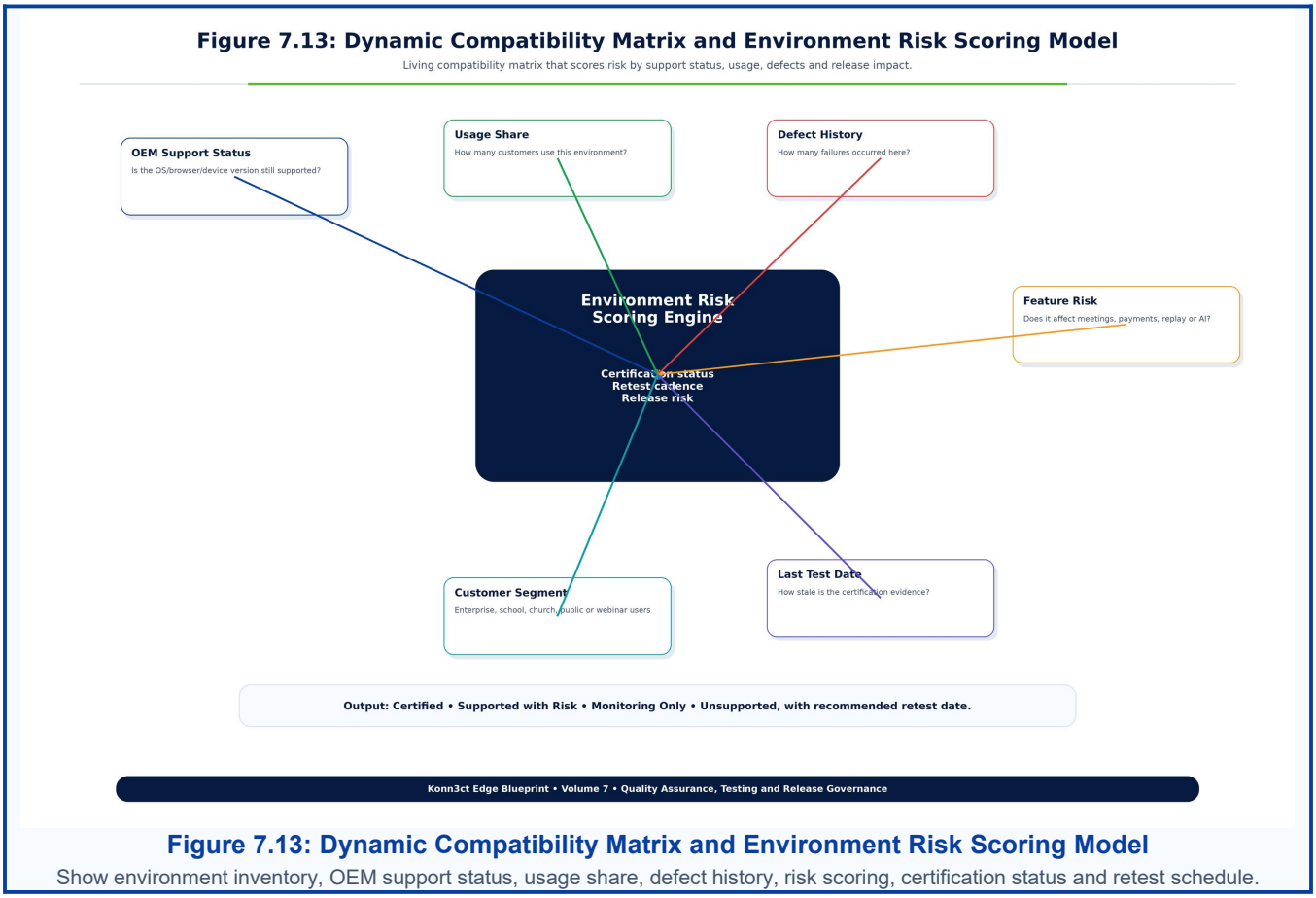


Figure 7.13: Dynamic Compatibility Matrix and Environment Risk Scoring Model

Show environment inventory, OEM support status, usage share, defect history, risk scoring, certification status and retest schedule.

Certification Status	Meaning	Action
Certified	Environment passed critical workflows and SLAs.	Use in supported environment list.
Supported with Risk	Environment mostly works but has known limitations.	Communicate limitations and retest regularly.
Monitoring Only	Environment is not primary but customer issues are tracked.	Do not prioritize unless usage increases.
Unsupported	Vendor support has ended or defects cannot be reasonably resolved.	Notify users and exclude from production SLA.

7.22 Layout and User Experience Quality Gate

Meeting layout quality must cover both mobile and web/desktop layouts. Konn3ct must avoid releasing meeting screens that are visually broken, confusing, unstable or hard to control.

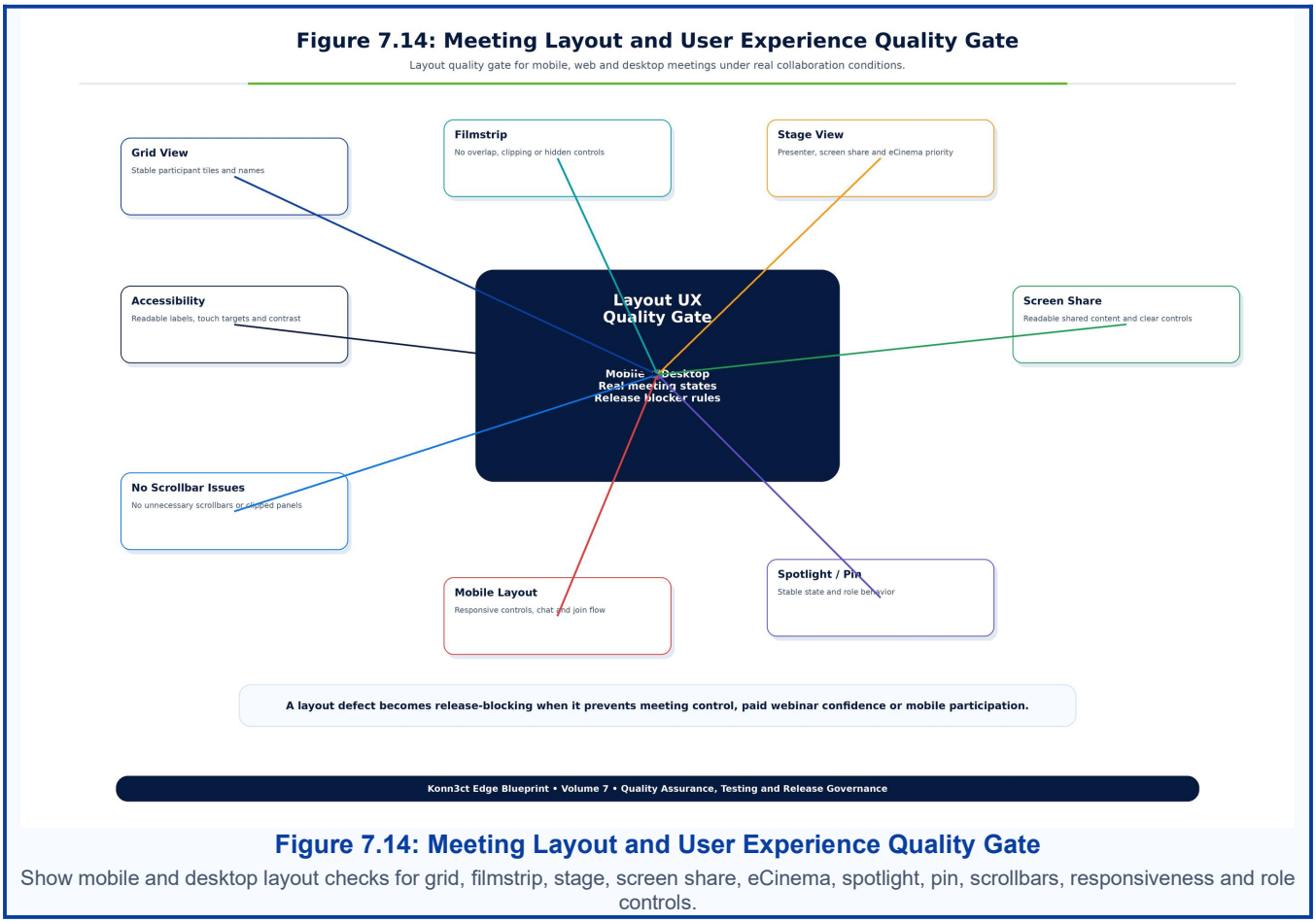


Figure 7.14: Meeting Layout and User Experience Quality Gate

Show mobile and desktop layout checks for grid, filmstrip, stage, screen share, eCinema, spotlight, pin, scrollbars, responsiveness and role controls.

- Stable grid layout for multiple participants.
- Filmstrip and stage view stability.
- Screen share and eCinema priority behavior.
- Spotlight and pin stability.
- No unnecessary scrollbars or clipped controls.
- Responsive behavior across desktop, tablet and mobile.
- Clear host and participant controls.
- Readable names, icons, chat, Q&A and AI panels.
- Layout behavior under recording, webinar and mobile join scenarios.

7.23 Device, Peripheral and Media Hardware Testing

Konn3ct depends on cameras, microphones, speakers, headsets, Bluetooth audio devices and screen-sharing behavior. QA should test representative device and peripheral combinations because many meeting failures originate from local hardware, permissions or device switching.

Hardware Area	Test Focus	Expected Result
Camera	Permission, camera selection, start/stop and switching.	Video works without freezing or wrong state.
Microphone	Permission, mute/unmute, device switching and noise behavior.	Audio is captured and state is clear.
Speaker/audio output	Output device selection and echo handling.	Users can hear clearly.
Headsets	Wired, USB and Bluetooth behavior.	Device changes do not break meeting state.
Screen sharing	Window, tab, full screen and browser-specific behavior.	Host/presenter can share predictably.
Low-end devices	CPU/memory pressure and battery behavior.	Graceful degradation where needed.

7.24 Performance, Load and Stress Testing

Performance testing must validate Konn3ct under realistic load. This includes API load, meeting concurrency, webinar attendance, payment activity, replay streaming, recording processing, AI job queues and mobile traffic.

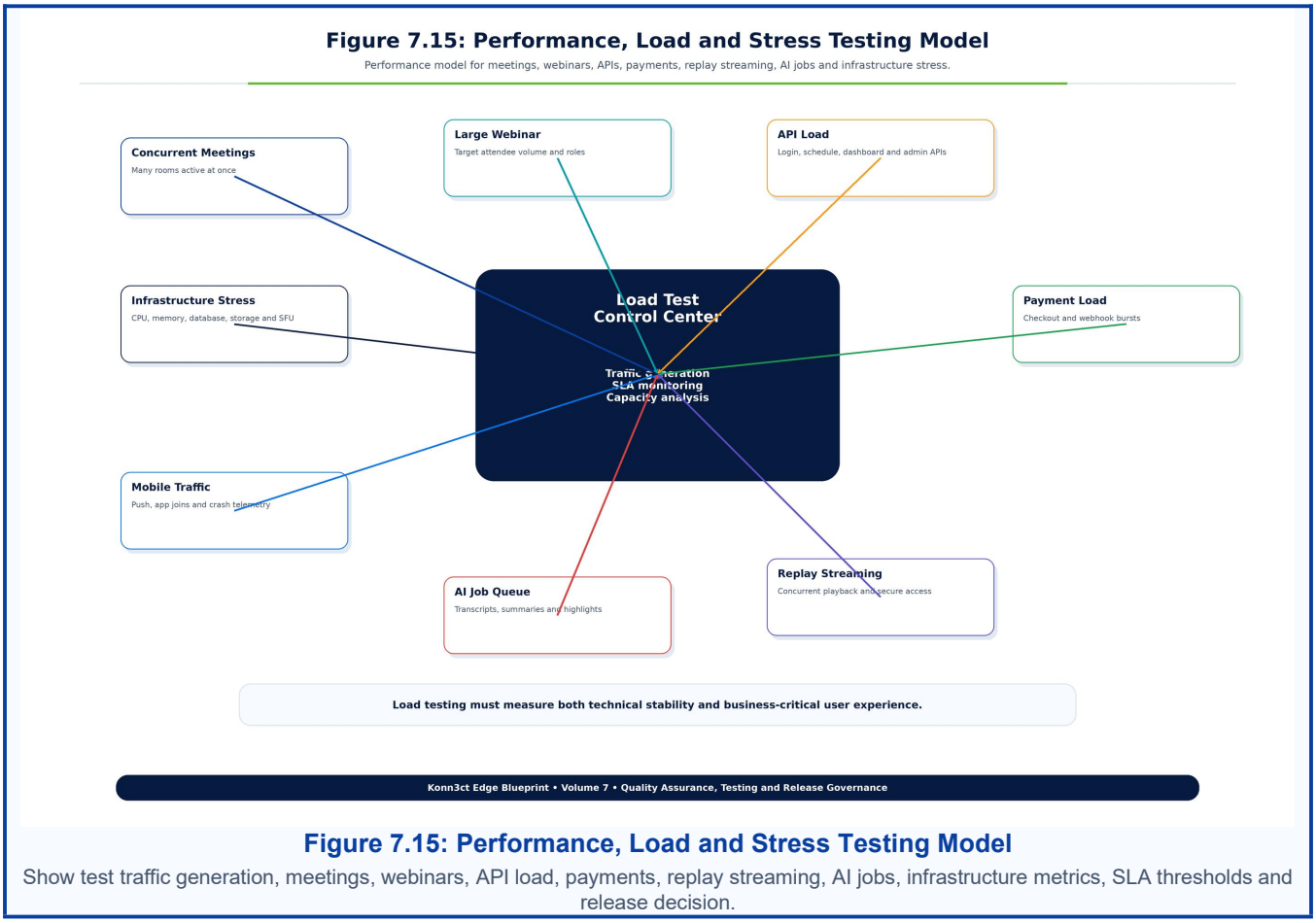


Figure 7.15: Performance, Load and Stress Testing Model

Show test traffic generation, meetings, webinars, API load, payments, replay streaming, AI jobs, infrastructure metrics, SLA thresholds and release decision.

Performance Area	What to Test	Measured Output
Concurrent meetings	Multiple rooms running at the same time.	Join success, SFU load and media quality.
Large webinars	Target attendee volume with host/presenter controls.	Stability, Q&A, attendance and recording.
API load	Login, scheduling, dashboard, payments and replay APIs.	Latency, error rate and database load.
Payment load	Many payment attempts and webhook callbacks.	Confirmation speed and entitlement accuracy.
Replay streaming	Concurrent playback and access checks.	Startup time, buffering and security.
AI processing	Summaries, transcripts and highlights in queue.	Completion time and cost impact.

7.25 Security and Privacy Testing

Security and privacy testing is release-blocking. Konn3ct must protect identities, meetings, payments, recordings, transcripts, AI outputs and private replays. Testing should validate access control, data protection and abuse prevention.

- Authentication and session management tests.
- Authorization and role-based access tests.

- Paid access bypass tests.
- Private and paid replay access tests.
- API input validation and rate-limit tests.
- Webhook verification tests.
- Secrets exposure and configuration review.
- Audit log verification.
- Recording and transcript privacy checks.
- Support access control and support override audit checks.

7.26 API and Integration Testing

API and integration testing should validate the interfaces used by web, mobile, payment gateways, Odoo, AI services, notifications and support systems. Integration failures can create customer-facing issues even where the user interface looks correct.

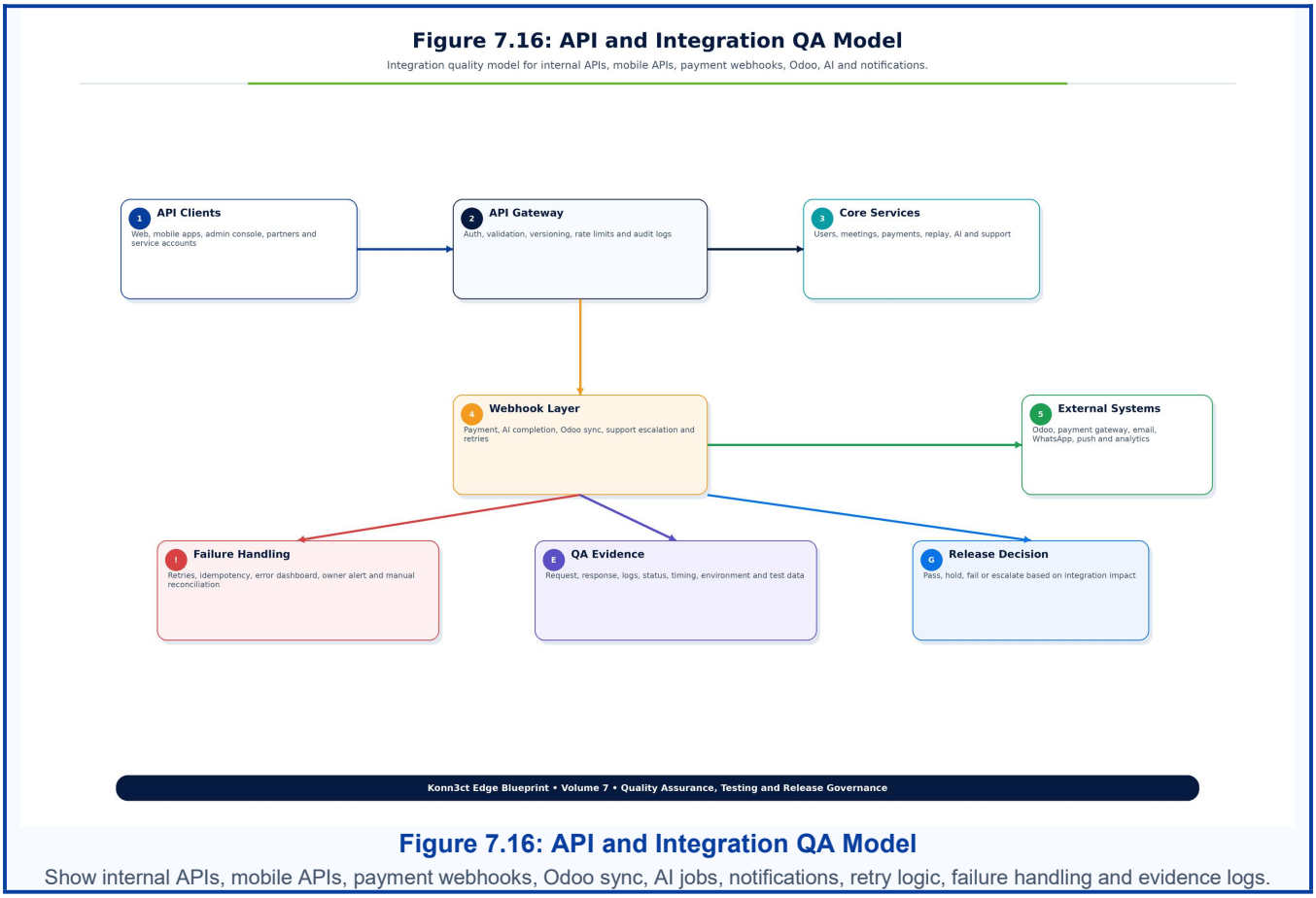


Figure 7.16: API and Integration QA Model

Show internal APIs, mobile APIs, payment webhooks, Odoo sync, AI jobs, notifications, retry logic, failure handling and evidence logs.

Integration Area	Test Focus	Failure Handling
Mobile APIs	Auth, meeting join, payment and replay endpoints.	Clear error states and logs.
Payment webhooks	Success, failure, duplicate and delayed callbacks.	Idempotency and entitlement accuracy.
Odoo sync	Leads, subscribers, payments, support tickets and customer health.	Retry queue and sync error dashboard.
AI jobs	Job creation, completion, failure and privacy rules.	Queue monitoring and user-friendly fallback.
Notifications	Email, WhatsApp, push and reminders.	Delivery logs and retry policy.
Support tools	Ticket creation, classification and escalation.	Traceability and root-cause tagging.

7.27 Odoo / CRM Integration QA

Odoo integration QA ensures that business workflows are not disconnected from platform usage. Konn3ct platform events should sync correctly into Odoo for leads, contacts, subscriptions, invoices, campaigns, helpdesk, renewals and customer success actions.

- New user registration creates or updates the right Odoo contact/lead.
- Subscription status is synchronized accurately.
- Paid event and paid replay revenue can be reported correctly.
- Payment failures trigger appropriate sales or support follow-up.
- Support tickets are classified and linked to customers.
- Customer health score receives usage, payment, support and engagement signals.
- Sync failures are logged, retried and visible to responsible teams.

7.28 Customer Support Feedback into QA

Customer support must feed the QA system. Support tickets, complaints, meeting failures, failed payment access, recording issues, replay access issues, mobile crashes and AI output complaints should become structured QA learning signals.

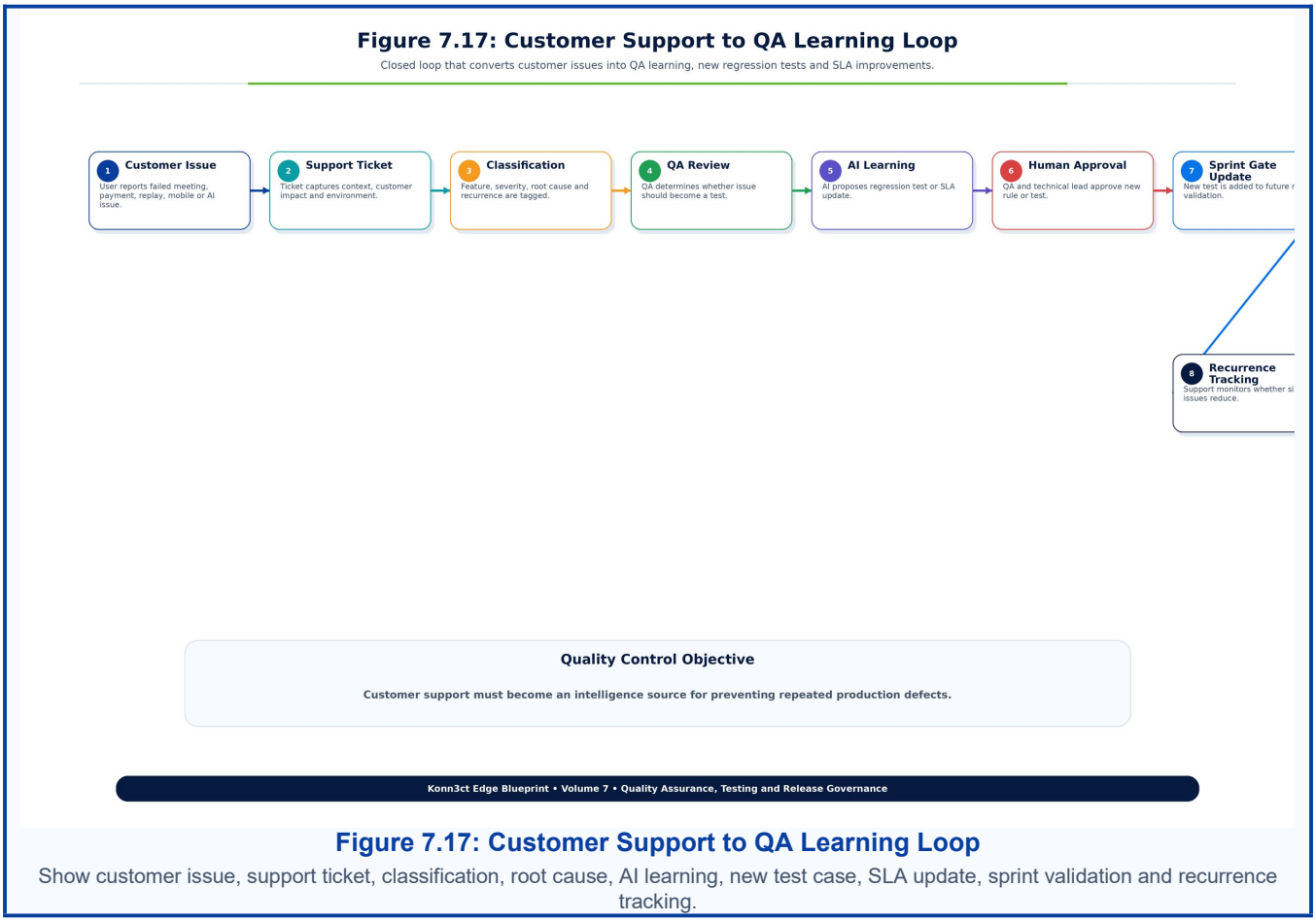


Figure 7.17: Customer Support to QA Learning Loop

Show customer issue, support ticket, classification, root cause, AI learning, new test case, SLA update, sprint validation and recurrence tracking.

Support Input	QA Learning Action	Expected Benefit
Cannot join meeting	Add browser/device/network scenario to regression tests.	Reduce repeated join failures.
Paid but no access	Test payment webhook and entitlement scenarios.	Protect paid user trust.
Replay unavailable	Test replay access, expiry and storage readiness.	Improve content reliability.
Mobile crash	Add device/OS/app version to compatibility matrix.	Improve mobile stability.

Support Input	QA Learning Action	Expected Benefit
AI summary missing	Test AI job queues and permission rules.	Improve AI feature trust.
Recurring confusion	Review UX wording, onboarding and support article.	Reduce avoidable tickets.

7.29 AI Developer Recommendation and Failed Component Advisory Model

When the AI QA engine detects a failure, it must do more than mark the test as failed. It should provide a technical and logical advisory to developers or the developer engine. This will reduce debugging time and improve sprint velocity.

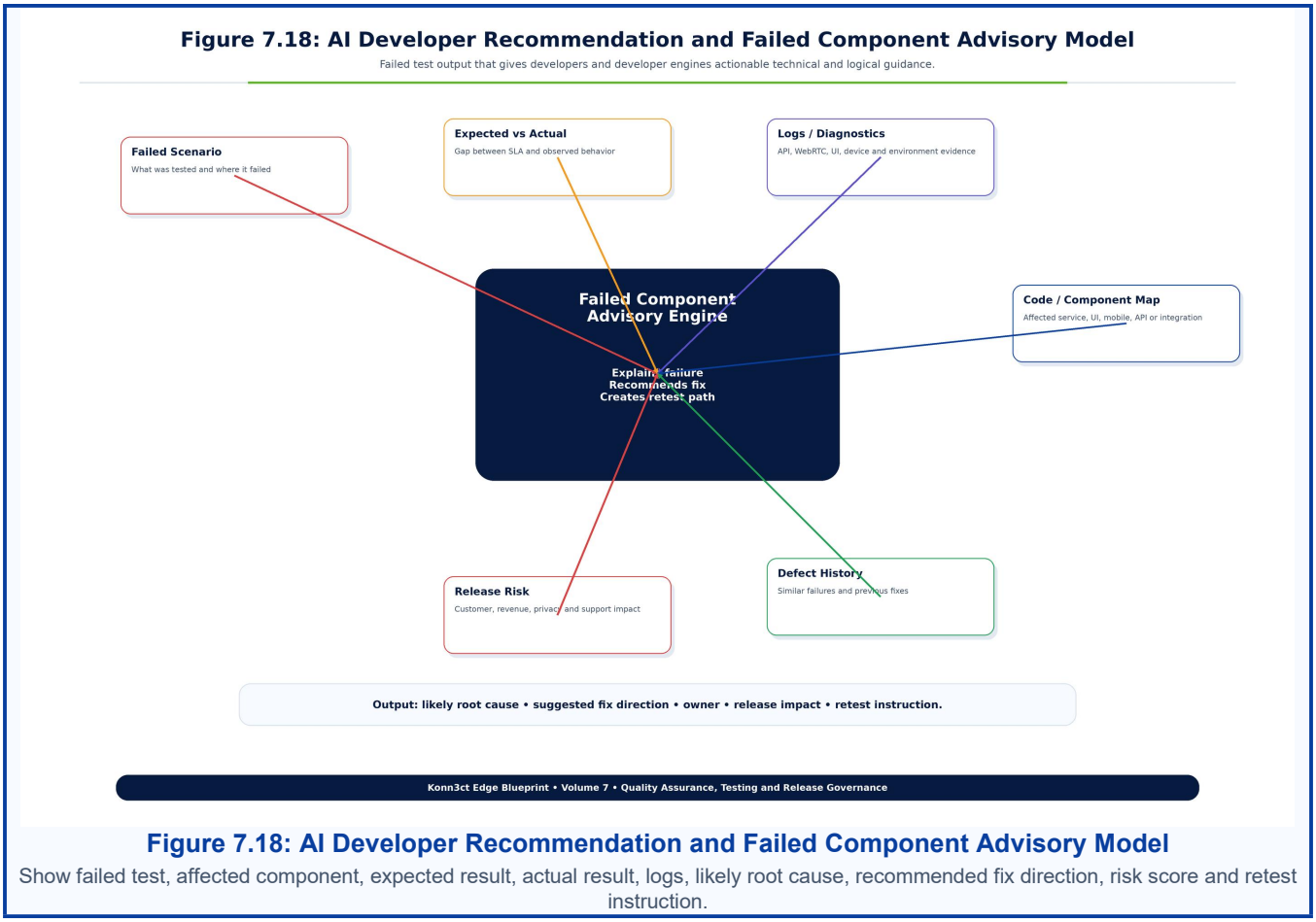


Figure 7.18: AI Developer Recommendation and Failed Component Advisory Model

Show failed test, affected component, expected result, actual result, logs, likely root cause, recommended fix direction, risk score and retest instruction.

Advisory Element	Description	Developer Value
Failed scenario summary	Plain explanation of what failed.	Quick understanding.
Expected vs actual result	Clear gap between requirement and observed behavior.	Reduces ambiguity.
Affected component	Frontend, backend, WebRTC, payment, replay, AI, Odoo or mobile.	Directs ownership.
Likely root cause	Reasoned hypothesis based on logs and history.	Speeds diagnosis.
Technical recommendation	Suggested fix direction or area to inspect.	Supports developer/developer engine.
Retest instruction	Scenario that must pass after fix.	Prevents incomplete resolution.
Release impact	Blocker, high, medium or low.	Guides prioritization.

7.30 Failed Test to Developer Fix Recommendation Workflow

The failed test workflow should be structured. Failures should move from detection to recommendation to ownership to fix to retest to regression learning.

1. AI QA engine detects a failed test or SLA breach.
2. Failure is mapped to affected subsystem and component owner.
3. Evidence is attached: screenshots, logs, metrics, diagnostics and test data.
4. AI generates logical and technical fix recommendation.
5. Developer or developer engine reviews and implements fix.
6. AI QA reruns affected tests and related regression tests.
7. Human QA validates user experience and edge cases.
8. Final fix evidence is attached to release readiness report.

7.31 Defect Classification and Severity Model

Defects should be classified consistently so that release decisions are disciplined. Severity should consider customer impact, revenue impact, privacy impact, security impact, operational impact and recurrence.

Severity	Definition	Release Decision
Critical	Breaks meeting join, payment access, private replay security, authentication, data privacy or production stability.	Release blocker.
High	Affects major user journey, paid customer experience, webinar reliability or mobile access.	Usually release blocker unless approved exception.
Medium	Affects secondary workflow with workaround.	May release only with documented risk.
Low	Cosmetic, wording or minor issue with no meaningful operational impact.	Can be scheduled after release.
Enhancement	Improvement request or usability suggestion.	Backlog prioritization.

7.32 Bug Triage and Resolution Governance

Bug triage should assign ownership, determine severity, confirm reproduction steps, evaluate release impact and define expected fix evidence. Triage must include QA, development and product; support should join where customer-facing issues are involved.

- Confirm whether the issue is reproducible.
- Identify affected environment, user role, feature and subsystem.
- Classify severity and release impact.
- Assign component owner and fix deadline.
- Require retest evidence before closure.
- Convert repeated defects into regression tests.
- Update customer support notes where customer communication is required.

7.33 Regression Testing Strategy

Regression testing ensures that fixed defects do not return. The regression suite should include core flows, historical defects, recurring support issues, production incidents, payment issues, replay access issues, mobile crashes and layout failures.

- Maintain permanent regression tests for critical workflows.
- Add tests from repeated support tickets.
- Add tests from every severe production incident.
- Run critical regression tests before every release.
- Run full regression before major product releases.

- Include both automated and human-observed regression tests.
- Track regression failures as higher risk than first-time failures.

7.34 Test Data and Test Environment Management

QA quality depends on reliable environments and safe test data. Konn3ct should maintain development, QA, staging, demo and load-testing environments with clear data rules. Production data should not be casually copied into test environments.

Environment	Purpose	QA Rule
Development	Developer testing and early checks.	Use synthetic or safe test data.
QA	Formal functional and automated testing.	Stable configuration and resettable test users.
Staging	Production-like release validation.	Mirror production configuration without exposing private data.
Demo	Sales and customer demonstration.	Use curated demo data and stable scenarios.
Load test	Performance and scale testing.	Isolate from production impact.
Production	Live users.	Only smoke tests and monitored post-release validation.

7.35 QA Evidence and Documentation Standard

Every release should produce evidence. QA evidence helps leadership trust the release decision and helps developers debug failures. Evidence should be stored in a structured release readiness report.

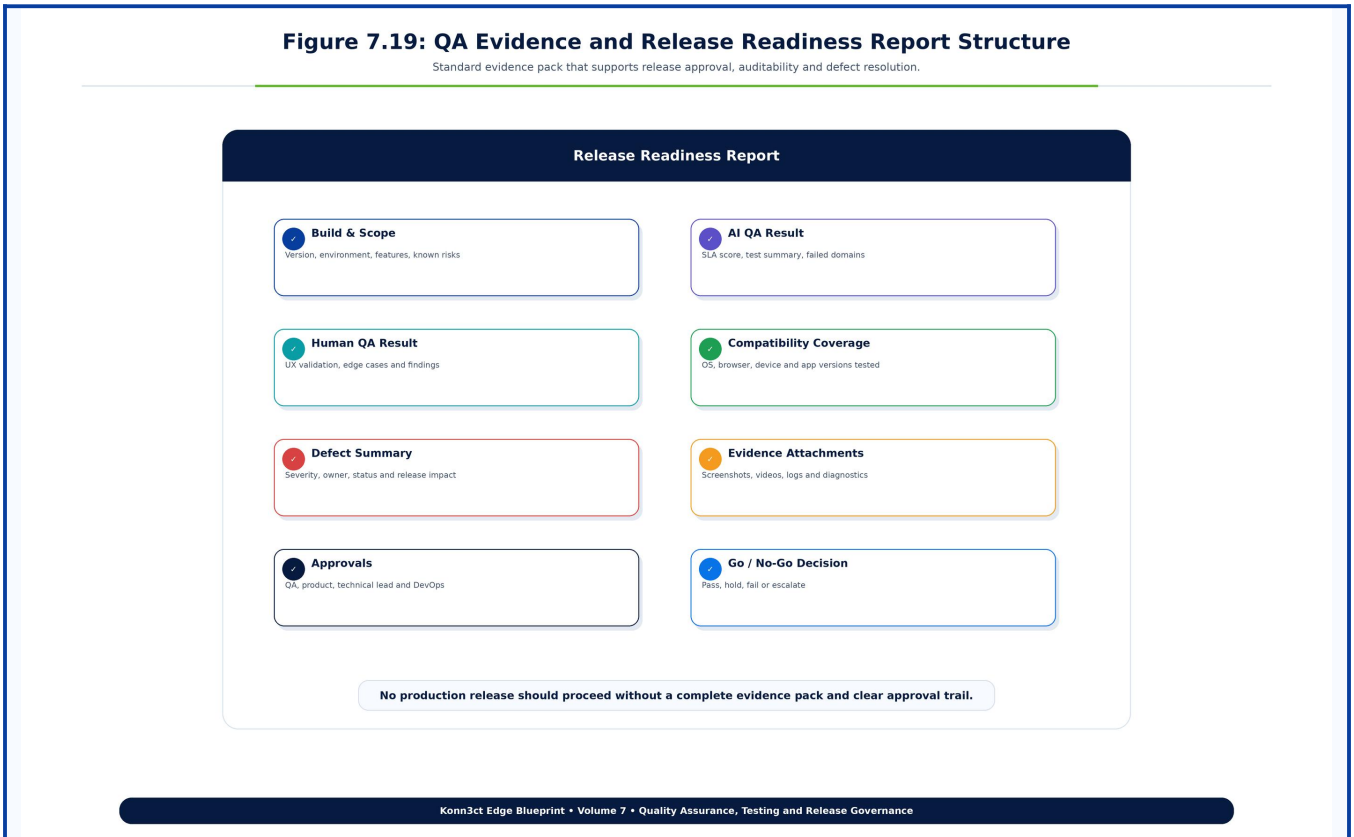


Figure 7.19: QA Evidence and Release Readiness Report Structure

Show test summary, SLA score, passed tests, failed tests, screenshots, videos, logs, diagnostics, known issues, approvals and release decision.

- Test plan and scope.
- Environment and build version.
- OS, browser, device and app version coverage.
- Automated AI QA report.
- Human QA summary.
- Screenshots, videos and diagnostics snapshots.
- Failed test list and developer recommendations.
- Known issues and risk acceptance.
- Product owner and technical lead approval.
- Post-release smoke test result.

7.36 QA Diagnostic Overlay Usage

The QA diagnostic overlay should help testers capture technical evidence from meetings and related flows. It should be available only to authorized QA or engineering users and should mask sensitive identifiers.

- Capture WebRTC state, producer/consumer counts and track state.
- Capture layout overflow, screen share priority and responsive behavior.
- Capture virtual background state and degradation.
- Capture device, browser and OS information.
- Capture issue snapshot export for engineering.
- Link snapshot to defect ticket and release evidence.
- Use diagnostic evidence to improve SLA rules and regression tests.

7.37 Production Monitoring and Post-Release QA

QA does not end at deployment. After production release, Konn3ct should run smoke tests, monitor dashboards, watch support tickets, track error rates and confirm that critical workflows remain healthy.

- Run immediate smoke tests after deployment.
- Monitor login, meeting join, payment, replay and AI job health.
- Track error spikes and support ticket increases.
- Confirm that release-specific features work in production.
- Define rollback signals and escalation path.
- Hold post-release review for major releases or incidents.

7.38 Release Blockers and Go/No-Go Criteria

The go/no-go decision must be explicit. Critical failures must block release regardless of schedule pressure. Release discipline is essential for customer trust.

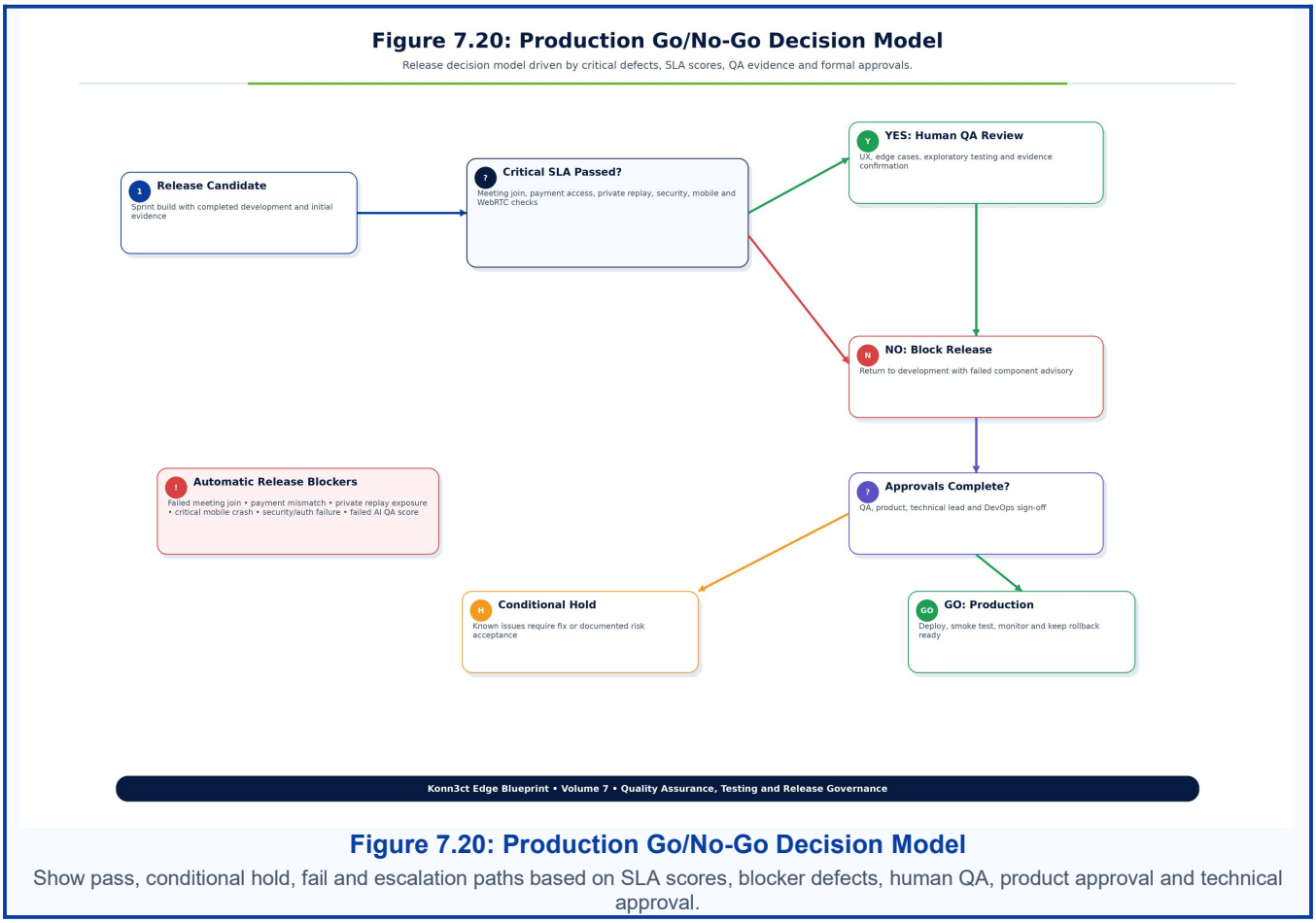


Figure 7.20: Production Go/No-Go Decision Model

Show pass, conditional hold, fail and escalation paths based on SLA scores, blocker defects, human QA, product approval and technical approval.

Release Blocker	Reason
Users cannot reliably join meetings on certified environments.	Core product failure.
Critical WebRTC audio/video or screen share failure.	Meeting trust failure.
Paid access is granted without verified payment.	Revenue and control failure.
Paid customer cannot access paid event or replay after successful payment.	Paid user trust failure.
Private or paid recordings are exposed through public unrestricted links.	Security and privacy failure.
Critical mobile crash affects certified devices.	Mobile adoption failure.
AI output exposes private data or bypasses access control.	Privacy and trust failure.
Security, authentication or authorization defect exists.	Enterprise trust failure.
AI QA engine fails critical sprint SLA.	Release readiness failure.

7.39 QA Roles and Responsibilities

QA requires clear ownership across teams. The QA lead owns the process, but product, development, support, DevOps and leadership all contribute to release quality.

Role	Responsibility
QA Lead	Owns QA strategy, test scope, evidence standard and release quality recommendation.
AI QA Owner	Manages AI QA engine, SLA library, automated tests and failed component advisory.
Manual QA Testers	Validate workflows, edge cases, UX, layout and real-world scenarios.
Developers	Fix defects, review AI recommendations and provide technical evidence.
Technical Lead	Approves technical readiness and release risk.
Product Owner	Approves business readiness and user experience acceptance.

Role	Responsibility
DevOps/SRE	Owns deployment readiness, monitoring, rollback and infrastructure health.
Support Lead	Feeds customer issues into QA and validates support learning loop.
Leadership	Enforces release discipline and prevents unsafe launch pressure.

7.40 QA KPIs and Quality SLAs

QA KPIs should make quality measurable. These KPIs should be reviewed regularly and connected to Free User growth, Paid User conversion, retention and support reduction.



Figure 7.21: QA KPI and SLA Dashboard

Show automated test pass rate, defect leakage, join success, payment access accuracy, recording readiness, mobile crash rate, support recurrence and release readiness score.

QA KPI	Meaning	Business Link
Automated test pass rate	Percentage of automated tests passing.	Improves release confidence.
AI QA release score	Composite score from SLA domains.	Blocks weak builds early.
Defect leakage	Defects found after release.	Measures release discipline.
Meeting join success	Users successfully joining meetings.	Protects activation and retention.
Payment access accuracy	Paid access matches payment status.	Protects revenue and trust.
Recording availability	Recordings ready within expected window.	Supports replay value.
Mobile crash rate	App stability across supported devices.	Protects mobile growth.
Support recurrence rate	Repeat support issues after resolution.	Measures learning effectiveness.

7.41 Guinness World Record QA and Evidence Validation

If Konn3ct pursues a future Guinness World Record attempt, QA must validate evidence readiness before the event. The attempt must be treated as a technical, operational, evidence and governance project, not only a marketing activity.



Figure 7.22: Guinness World Record QA Evidence Validation Model

Show registration validation, attendance logs, timestamps, full recording, server metrics, witness evidence, audit logs and evidence submission pack.

- Validate participant registration and unique attendance tracking.
- Validate join, leave and duration records.
- Validate full-session recording continuity.
- Validate server metrics, uptime and incident timeline.
- Validate witness and independent audit evidence support.
- Validate exportable attendance and evidence pack.
- Validate no manual manipulation of participant numbers.
- Validate backup plan for recording, logs and monitoring.

7.42 QA Roadmap: 90 Days, 6 Months and 12 Months

QA maturity should be built in phases. The first phase should stabilize the basics and define release discipline. The second phase should deepen automation, compatibility and support learning. The third phase should mature the independent AI QA engine and self-learning SLA system.



Figure 7.23: QA Roadmap: 90 Days, 6 Months and 12 Months

Show phased QA maturity across core testing, AI QA engine, SLA library, compatibility matrix, support learning, production monitoring and international readiness.

Phase	QA Priorities	Expected Outcome
0-90 Days	Define SLA library, core regression suite, compatibility matrix, release blockers, QA evidence standard and AI QA MVP.	Basic quality discipline and release gate foundation.
3-6 Months	Expand AI QA coverage, mobile/browser certification, flexible scenario engine, support learning loop and integration testing.	Broader automated coverage and improved customer issue prevention.
6-12 Months	Mature self-learning SLA management, developer advisory engine, large-scale testing, country-specific matrix and executive quality dashboards.	Enterprise-grade release governance and international readiness.

7.43 QA Risks and Mitigation Plan

The QA model must manage known risks that can weaken production readiness.

Risk	Impact	Mitigation
Weak AI QA rules	False confidence or excessive false failures.	Human approval, continuous calibration and evidence review.
Incomplete compatibility matrix	Users fail on untested OS, devices or browsers.	OEM-supported matrix and usage-based risk scoring.
Poor test data	Tests pass but real users fail.	Realistic test personas, event data and payment scenarios.
Unstable staging	QA cannot trust results.	Production-like staging and environment governance.
Repeated defects	Customer trust and support cost worsen.	Regression suite and support learning loop.
Payment defects	Revenue loss and paid user frustration.	Payment-specific release blockers and entitlement tests.
Replay privacy failure	Trust and compliance risk.	Private access testing and storage policy checks.
Release pressure	Unsafe releases reach production.	Go/no-go authority and leadership discipline.

7.44 Volume 7 Conclusion

Volume 7 establishes Konn3ct QA as an intelligent, evidence-driven and governance-controlled system. The platform must not depend on informal confidence or last-minute manual testing. Quality must be planned, automated, measured, reviewed and approved.

The Independent AI QA Engine, intelligent SLA library, OEM-supported compatibility matrix, flexible scenario engine, support learning loop and developer recommendation model together create a stronger release discipline. They ensure that each sprint is tested not only for what was built, but also for how customers may actually use it.

Final QA principle: Konn3ct must release only when quality evidence supports the decision: AI QA has tested it, SLAs have been met, human QA has validated it, product has accepted it, technical leadership has approved it and production monitoring is ready.